

$$\Theta' \in \bar{\mathcal{W}}_{\infty}^{\infty} = \mathcal{Y}' \cap \bar{\mathcal{W}} = \mathcal{P} \in \mathcal{C} \cap \bar{\mathcal{W}}' \cap \hat{\mathcal{B}}^{\infty} = \hat{\mathcal{Y}} \cap \bar{\mathcal{W}}$$